

- This exam contains 8 pages (including the front and back of this cover page) and 8 problems. Check to see if any pages are missing.
- This is the first exam and will count for 20% of your final grade. You have 105 minutes (5:15 pm - 7:00 pm) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- Every problem on this exam, other than true-or-false problems, requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 220 points total in this exam, as well as 1 bonus questions worth a total of 20 possible points.

Page:	3	4	5	6	Total
Points:	20	60	60	80	220
Score:					

Bonus:	
--------	--

1. Mark each statement as ☐ True or ☐ False by completely shading in the box corresponding to your answer.

For readability in the key, correct answers are colored in red, e.g. **Correct Answer**, while incorrect answers are marked with a strike, e.g. ~~Incorrect Answer~~.

- (a) ☐ 2 points Elementary row operation on an augmented matrix never change the solution set of the associated linear system.

☐ True☐ False

- (b) ☐ 2 points The reduced row echelon form of a matrix is unique.

☐ True☐ False

- (c) ☐ 2 points A homogeneous system is always consistent.

☐ True☐ False

- (d) ☐ 2 points If A , B , and C are matrices such that $AB = AC$, then $B = C$.

☐ True☐ False

- (e) ☐ 2 points The columns of any 4×5 matrix are linearly dependent.

☐ True☐ False

- (f) ☐ 2 points Every matrix transformation is a linear transformation.

☐ True☐ False

- (g) ☐ 2 points A linear transformation preserves the operations of vector addition and scalar multiplication.

☐ True☐ False

- (h) ☐ 2 points If two vectors are linearly independent, then they both lie on the same line through the origin.

☐ True☐ False

- (i) ☐ 2 points If A and B are $n \times n$ matrices such that $AB = I$, then A and B are invertible.

☐ True☐ False

- (j) ☐ 2 points A column of the standard matrix for a linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the image of a column of the $n \times n$ identity matrix.

☐ True☐ False

2. 30 points Determine the value(s) of $h \in \mathbb{R}$ such that the following matrix is the augmented matrix of a consistent linear system:

$$\begin{bmatrix} 2 & 3 & \cos(h) \\ 4 & 6 & \sqrt{2} \end{bmatrix}$$

3. 30 points Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 1 \end{bmatrix}$. Do the columns of A^T span $\mathbb{R}^2$? Justify your answer.

4. 30 points Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be a linear transformation such that

$$T(x_1, x_2) = \begin{bmatrix} x_1 - 2x_2 \\ -x_1 + 3x_2 \\ 3x_1 - 2x_2 \end{bmatrix}.$$

Find the standard matrix corresponding to the linear transformation T .

5. 30 points Let $A = \begin{bmatrix} 3 & 2 \\ 7 & 4 \end{bmatrix}$. Give an explicit formula for a general solution for the inhomogeneous system $Ax = b$ without using back-substitution or row-reduction.

6. 40 points Suppose $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a one-to-one (i.e. T is injective) linear transformation. What relationship between n and m must be true?

7. Let A be an $n \times n$ matrix, and let the linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be defined by $T(x) = Ax$. Suppose $T(u) = T(v)$ for some $u, v \in \mathbb{R}^n$ with $u \neq v$.

(a) 20 points Does the equation $Ax = 0$ have a nontrivial solution? Justify your answer.

(b) 20 points Does the equation $Ax = z$ have a unique solution for any $z \in \mathbb{R}^n$? Justify your answer.

8. 20 points (bonus) Recall that a pair of complex numbers $z_1 = x_1 + ix_2$ and $z_2 = y_1 + iy_2$ satisfy

$$z_1 z_2 = (x_1 + ix_2)(y_1 + iy_2) = (x_1 y_1 - x_2 y_2) + (x_1 y_2 + x_2 y_1)i.$$

Let $\mathbb{C}$ denote the complex numbers and let $\mathbb{M}_2(\mathbb{R})$ denote the set of 2×2 matrices with real entries. To solve this problem, find a pair of 2×2 matrices A and B such that, for any complex numbers $z_1, z_2 \in \mathbb{C}$, the map $T : \mathbb{C} \rightarrow \mathbb{M}_2(\mathbb{R})$ defined by $T(z) = T(a + ib) = aA + bB$ satisfies

$$T(z_1 z_2) = T(z_1)T(z_2)$$

Since T is linear, this shows that the complex numbers have a representation as matrices, and that this representation preserves both the additive and multiplicative structure of the complex numbers.

HINT: Try using a matrix B of the form $B = \begin{bmatrix} b_1 & -b_2 \\ b_2 & b_1 \end{bmatrix}$ for some real numbers b_1 and b_2 .

NOTE: There was a minor typo in this problem (see the remark after the answer given below)

